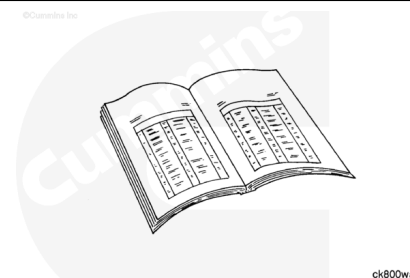


Trainings ▾

Preparatory Steps

- Remove the air piping from the intake manifold.
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3.
(/qs3/pubsys2/xml/en/procedures/35/35-003-011-tr.html)
- Remove the engine brakes, if equipped. Refer to Procedure 020-024 in Section 20.
(/qs3/pubsys2/xml/en/procedures/35/35-020-024-tr.html)
- Remove the push tubes and push rods. Refer to Procedure 004-014 in Section 4.
(/qs3/pubsys2/xml/en/procedures/35/35-004-014-tr.html)
- Remove the gear cover. Refer to Procedure 001-031 in Section 1. (/qs3/pubsys2/xml/en/procedures/35/35-001-031.html)



Note : It is possible to remove the camshaft without removing the cylinder head. The cam followers can be tied up and away from the camshaft with wire.

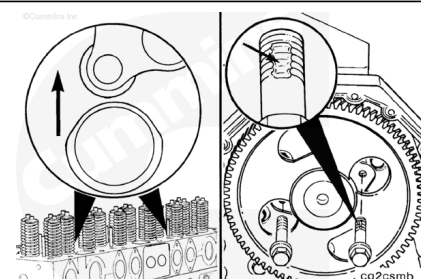
- Remove the camshaft idler gear. Refer to Procedure 001-036 in Section 1.
(/qs3/pubsys2/xml/en/procedures/35/35-001-036-tr.html)

Remove

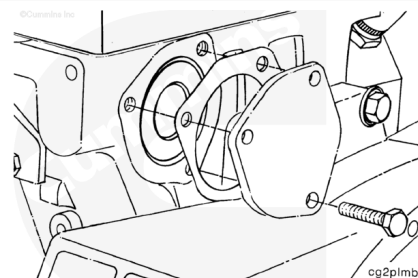
Raise the cam follower assemblies, and use a wire to tie the cam followers up away from the camshaft.

Rotate the camshaft to align the holes in the camshaft gear with the thrust plate capscrews.

Remove the thrust plate capscrews.

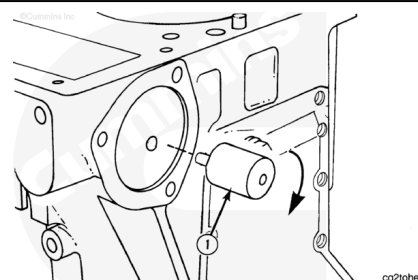


Remove the three capscrews and the camshaft rear cover plate.

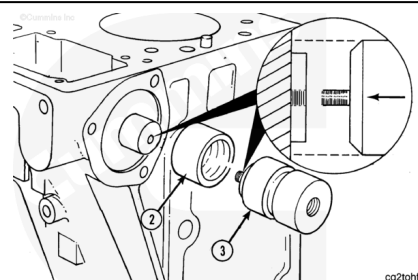


Install camshaft installation pilot, Part Number 3376388, as follows:

- Install the expander (1), Part Number 3376923, in the camshaft through the rear cover plate opening.
- Turn the screw inside the expander **clockwise** to expand the swell plug.

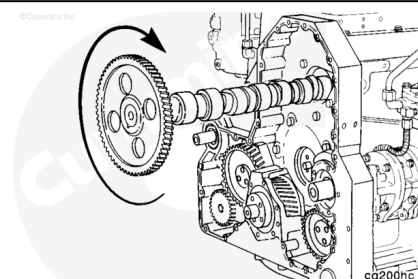


- Install the arbor sleeve (2) over the expander.
- Install the locating pilot extensions (3) to the expander.



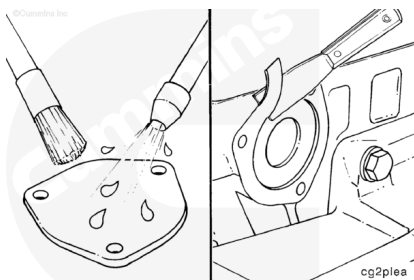
Use one hand to rotate slowly and pull the camshaft from the cylinder block and the other hand to balance the camshaft as it is removed.

Remove the camshaft pilot tool.



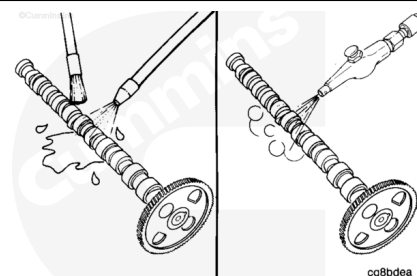
Clean and Inspect for Reuse

Clean the camshaft rear cover plate and the cylinder block mating surface.



⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



⚠ WARNING ⚠

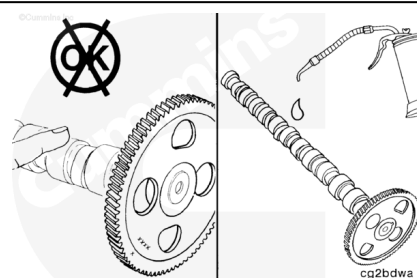
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the camshaft with steam or solvent.

Dry with compressed air.

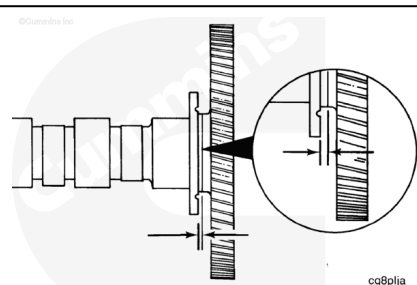
⚠ CAUTION ⚠

After the camshaft has been steam cleaned, do not touch the machined surfaces with bare hands. This will cause rust to form, which will damage the camshaft. Lubricate the camshaft with clean 15W-40 oil before handling.



Measure the clearance between the gear hub and the thrust plate.

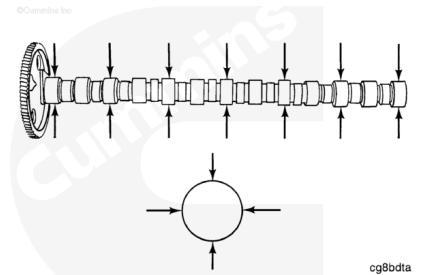
Camshaft Thrust Plate Clearance		
mm		in
0.13	MIN	0.005
0.33	MAX	0.013



Measure the outside diameter of the seven camshaft bushing journals.

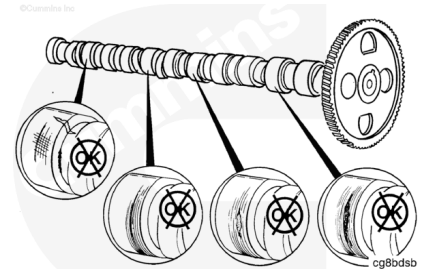
Camshaft Bushing Journal Outside Diameter

mm		in
71.960	MIN	2.8331
72.013	MAX	2.8352

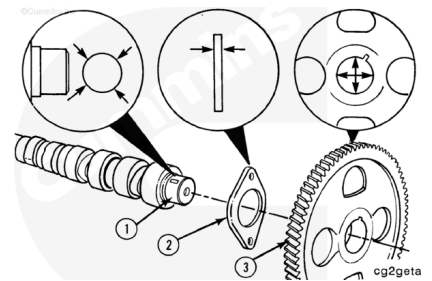


Inspect the camshaft lobes for cracks, scratches, or other damage.

Reference Camshaft Reuse Guidelines for engines with crowned rollers, Bulletin 3666052 (</qs3/pubsys2/xml/en/bulletin/3666052.html>).



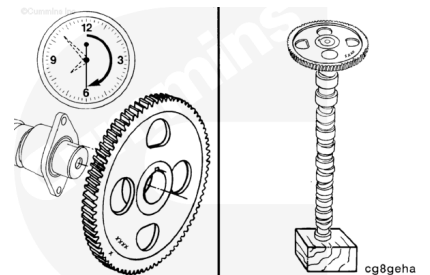
Remove the camshaft gear, and measure the camshaft nose (1), thrust plate (2), and gear (3). Refer to Procedure 001-013 in Section 1. (</qs3/pubsys2/xml/en/procedures/35/35-001-013.html>)



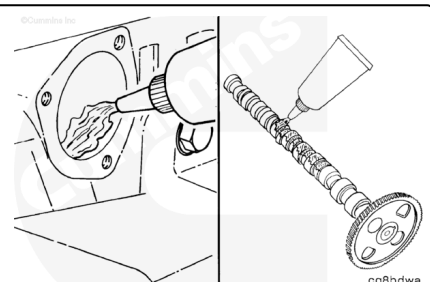
Install

Install the camshaft gear on the camshaft. Refer to Procedure 001-013 in Section 1.

(</qs3/pubsys2/xml/en/procedures/35/35-001-013.html>)

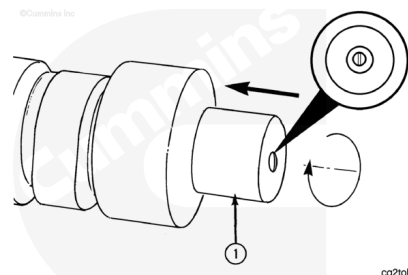


Use Lubriplate™ 105 or equivalent, to coat the camshaft bushings and camshaft.

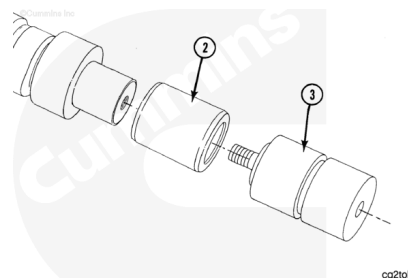


Install camshaft installation pilot, Part Number 3376388, as follows:

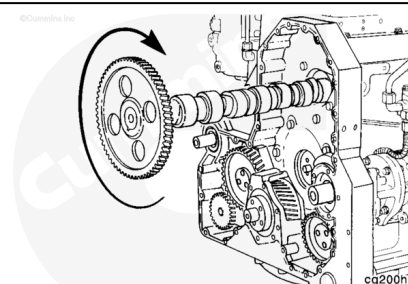
- Install expander (1), Part Number 3376923,
- Turn the screw inside the expander **clockwise** to expand the swell plug.



- Install the arbor sleeve (2) over the expander.
- Install the locating pilot extensions (3) to the expander.

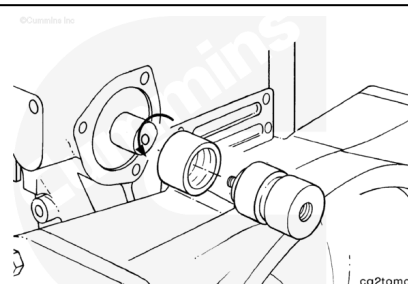


Rotate the camshaft slowly as it is being installed in the cylinder block.



The camshaft installation pilot **must** be disassembled as it clears the number seven camshaft bushing to prevent interference with the rib across the top of die-cast flywheel housings.

Remove the camshaft pilot.

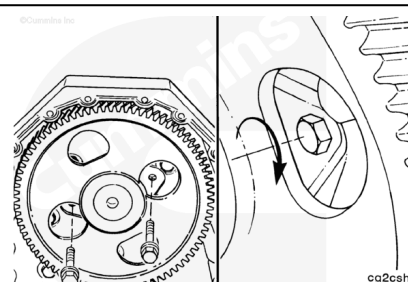


Rotate the camshaft to align the holes in the camshaft gear with the thrust plate capscrew holes in the cylinder block.

Align the capscrew holes in the thrust plate with the cylinder block capscrew holes.

Install the capscrews.

Torque Value: 47 n•m [35 ft-lb]



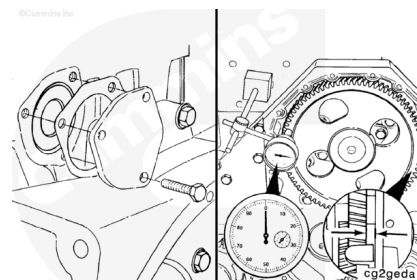
Use a new gasket to install the camshaft rear cover plate.

Tighten the three capscrews.

Torque Value: 47 n•m [35 ft-lb]

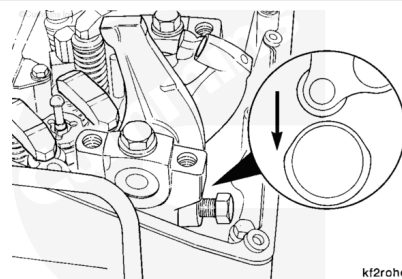
Measure the camshaft end clearance.

Camshaft End Clearance		
mm		in
0.13	MIN	0.005
0.33	MAX	0.013



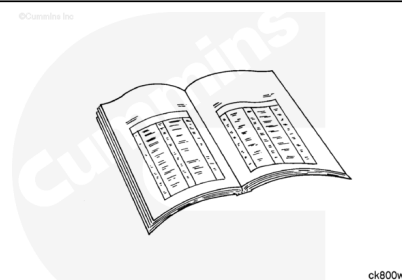
Do **not** drop the cam follower levers on the camshaft lobes.

Carefully lower the cam followers onto the camshaft.



Finishing Steps

- Install the camshaft idler gear. Refer to Procedure 001-036 in Section 1.
(/qs3/pubsys2/xml/en/procedures/35/35-001-036-tr.html)
- Check the engine timing. Refer to Procedure 006-025 in Section 6. (/qs3/pubsys2/xml/en/procedures/35/35-006-025-tr.html)
- Install the gear cover. Refer to Procedure 001-031 in Section 1. (/qs3/pubsys2/xml/en/procedures/35/35-001-031.html)
- Install the push tubes and push rods. Refer to Procedure 004-014 in Section 4.
(/qs3/pubsys2/xml/en/procedures/35/35-004-014-tr.html)
- Adjust the valves and injectors. Refer to Procedure 003-004 in Section 3.
(/qs3/pubsys2/xml/en/procedures/35/35-003-004-tr.html)
- Install the engine brakes, if equipped. Refer to Procedure 020-024 in Section 20.



(/qs3/pubsys2/xml/en/procedures/35/35-020-024-tr.html)

- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/35/35-003-011-tr.html)
- Operate the engine to normal operating temperature and check for leaks.

Last Modified: 21-Nov-2011
